PathwayAI: Making the Hidden Patient Journey Visible—and Actionable

HOW LYME DISEASE BURDEN CAN CHANGE

Same-study comparison: Hook et al. (2022)

Confirmed localized Lyme disease: mean societal cost USD 1,307, in 2016 dollars.

Confirmed disseminated Lyme disease: mean societal cost USD 3,251, in 2016 dollars.

Societal costs include medical, nonmedical and productivity components; do not add these components again.

Study averages are not individual expected bills. Disseminated disease and delayed treatment are not interchangeable labels. The group difference does not establish causal savings from earlier treatment.

Tick bite without illness: expenses vary; no fixed cost is assigned.

Horn et al. (2025): 55 of 253 returning participants reported ongoing symptoms around three months; 466 cases enrolled. Selected follow-up participants, not a disability probability.

Adrion et al. (2015): within Lyme cases, symptom-related diagnosis codes were associated with USD 3,798 higher adjusted medical costs over 12 months. Historical commercially insured population; allowed amounts, not patient bills. Do not add to Hook societal costs.

Johnson et al. (2014): selected persistent-illness respondents reported work and functional impacts. Survey percentages are not probabilities of disability among newly diagnosed cases.

The situations are not a progression model. No transition probabilities, county disability count or single long-term cost is estimated.

PRACTICAL ACTIONS

Situation: Planning time outdoors | PathwayAI offers: Prevention information and a downloadable preparation plan | Community opportunity: Share prevention education through websites and local events

Situation: Tick found; illness may or may not develop | PathwayAI offers: Removal information and guidance on contacting a clinician | Community opportunity: Make reliable tick-bite information easy to reach

Situation: Seeking evaluation or treatment | PathwayAI offers: Care resources, mixed-language notes and a reviewed visit summary | Community opportunity: Support affordable-care navigation and interpreter access

Situation: Ongoing symptoms or activity limitations | PathwayAI offers: Dated records of daily function, work changes and caregiving | Community opportunity: Connect residents with follow-up and practical support

Situation: Longer-term support needs | PathwayAI offers: Organized records and benefits-preparation resources | Community opportunity: Connect residents with qualified benefits advisers and existing services

SOURCES

https://wwwnc.cdc.gov/eid/article/28/6/21-1335_article

https://doi.org/10.3389/fmed.2025.1577936

https://doi.org/10.1371/journal.pone.0116767

https://doi.org/10.7717/peerj.322